

Sai Nilohita Reddy Male

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SUMMARY

- Graduate Computer Science student with a Concentration in Applied Artificial Intelligence at the University of Ottawa.
- Research Assistant focused on applying reinforcement learning to enhance online learning models and improve adaptability in dynamic data environments.
- Proficient in several programming languages, with experience in GenAI frameworks, data analysis, and Large Language Model (LLM) applications.
- Experienced in developing and evaluating machine learning and reinforcement learning models for applied research.
- Teaching Assistant for undergraduate computer science and math courses.
- Collaborated with various teams on research and development projects.

EDUCATION

Master of Computer Science (Thesis, CO-OP)

University of Ottawa

Concentration in Applied Artificial Intelligence

Sept 2025 – Present

Ottawa, Ontario

Bachelor of Computer Science (Honours)

Carleton University

Minors: Statistics and Mathematics – CGPA (12-Point): 10.32

Sept 2021 – Apr 2025

Ottawa, Ontario

TECHNICAL SKILLS

Programming Languages: Python, C/C++, R, Java, SQL, HTML, CSS, JavaScript, Haskell

Data Science and ML: PyTorch, scikit-learn, Pandas, NumPy, Matplotlib, OpenCV, Jupyter, Seaborn

Development: Git, RESTful APIs, Node.js, React, MongoDB, CMake, Qt Creator

WORK EXPERIENCE

Research Assistant

University of Ottawa

Sept 2025 – Present

Ottawa, Ontario

- Conducting research on applying reinforcement learning (RL) to adapt online learning (OL) models under feature drift conditions.
- Implementing baseline online learners (Hoeffding Tree and Hoeffding Adaptive Tree) using scikit-multiflow and River frameworks.
- Running experiments on benchmark data streams (e.g., Electricity dataset) and analyzing model stability and adaptability.

Teaching Assistant

University of Ottawa

Sept 2025 – Dec 2025

Ottawa, Ontario

- Conducting tutorial and lab sessions on advanced C++ topics, including templates, RAII, and design patterns, for over 100 undergraduate students.
- Developing tutorial materials that include code examples to reinforce lecture concepts.
- Designing, managing, and grading assignments using GitHub Classroom, CMake, and continuous integration workflows (automated grading).

Teaching Assistant

Carleton University

Jan 2023 – Apr 2023

Ottawa, Ontario

- Conducted tutorials and office hours for Discrete Mathematics, covering topics such as algorithmic complexity, propositional logic, and set theory.
- Created an engaging learning environment for students, strengthening their problem-solving skills.
- Developed an efficient grading rubric, reducing grading time while ensuring constructive feedback.

PROJECTS

Analyzing the Integration of Ethical Principles in AI/ML Workflows

- Built a large-scale data mining and classification (using LLMs: Llama 3.3 70B, DeepSeek R1) pipeline to extract and preprocess 252K+ closed GitHub issues from 28 AI/ML repositories.
- Performed model evaluation and analysis, identifying class imbalance in LLM predictions.
- Designed a survey to assess developers' awareness and application of ethical principles in real-world workflows to promote ethical AI.

Wordle Solver

- Optimized the **SARSA** reinforcement learning algorithm for solving Wordle, achieving 3.58 average guesses per game, outperforming human players by 15%.
- Built the environment using **Gymnasium**, providing seamless integration with reinforcement learning for efficient training and evaluation.
- Collaborated with a team to perform a comparative analysis of 5 reinforcement learning algorithms on 2,309 unique Wordle puzzles, utilizing **NumPy** and **Pandas** for data analysis and processing.

RaDoTech Device Simulation

- Collaborated with a team to simulate a health monitoring device using **Qt** and **C++**, ensuring successful functionality across 11 rigorous test scenarios.
- Implemented **unit testing** to validate functionality and maintain system reliability.
- Applied design patterns such as **Model-View-Controller**, **State**, and **Observer** to build a scalable and maintainable system architecture.

Open Gallery

- Developed a secure web application for artists to showcase their work, featuring artwork browsing, reviews, and workshop enrollment for patrons and artists.
- Designed a scalable **MongoDB** database to efficiently manage user accounts, artwork data, and interactions across multiple collections.
- Engineered and tested **RESTful API** endpoints to support account creation, artwork interactions, notifications, and data retrieval, ensuring robust **CRUD** operations.

Customer Complaint Prediction using Supervised Learning

- Built a classification pipeline on the Customer Personality Analysis dataset by implementing feature engineering and hyperparameter tuning across six algorithms.
- Applied undersampling and SMOTE oversampling to address class imbalance and generated confusion matrices and ROC curves for evaluation.

VOLUNTEER EXPERIENCE

Medical Volunteer

Satya Sai Centre

Jan 2020 – Aug 2021

Gaborone, Botswana

- Assisted patients with vital sign checks and basic care in a calm and compassionate manner.
- Accurately prepared and dispensed medications and eyeglasses based on prescriptions.
- Worked closely with clinic staff to keep daily operations organized and patient flow efficient.

OTHER EXPERIENCE

Tutoring

- Helped IGCSE and A-Level students understand core Computer Science topics through one-on-one summer tutoring sessions.
- Explained how to write pseudocode and implement basic programs using Visual Basic.
Supported students in improving problem-solving and debugging skills through practice exercises.

RELEVANT COURSEWORK

Machine Learning: Deep Learning and Reinforcement Learning, Computer Vision, Algorithms for Data Science, AI for Cybersecurity Applications

Software Development: Object-Oriented Software Engineering, Operating Systems, Web Development